

### 3.1 Sub-module 1: Getting Familiar with Machine Learning (ML)

This sub-module teaches students to train simplified ML models to create design artefacts {ML-Obj}. Students develop procedural knowledge through practical guided activities (e.g. RunwayML step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating a logo). Students are first provided with a basic literacy of ML including technical vocabulary {ML-Act1} and primary functionalities {ML-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of ML tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of ML within the design practice {ML-Act3}.

At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial {ML-Act4.3} on how to use an ML tool (e.g. runway) starting from setting a design goal {ML-Act4.1} and creating their dataset {ML-Act4.2}. Then, to familiarise themselves with the procedure, students individually apply what they learned by training an ML model aligned with the design goal {ML-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results {ML-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {ML-Act7}.

Educational Objective {ML-Obj} :

Students will learn to train a Machine Learning model to create design artifacts.

| Sub-module 1:<br>Getting familiar<br>w/ ML | The cognitive process dimension |                                         |                        |                |                 |               |
|--------------------------------------------|---------------------------------|-----------------------------------------|------------------------|----------------|-----------------|---------------|
|                                            | (1)<br>Remember                 | (2)<br>Understand                       | (3)<br>Apply           | (4)<br>Analyze | (5)<br>Evaluate | (6)<br>Create |
| (a) Factual<br>Knowledge                   |                                 | {ML-Act.1}                              |                        |                |                 |               |
| (b) Conceptual<br>Knowledge                |                                 | {ML-Act.2}<br>{ML-Act.3}                |                        |                |                 |               |
| (c) Procedural<br>Knowledge                |                                 | {ML-Act.4}<br>{ML-Act.4.1, 4.2,<br>4.3} | {ML-Obj}<br>{ML-Act.5} |                |                 |               |
| (d) Meta<br>Knowledge                      |                                 |                                         |                        | {ML-Act.6}     | {ML-Act.7}      |               |

{ML-Act1} Activities intended to provide students with vocabulary knowledge of ML;

{ML-Act2} Activities intended to provide students with a basic functionality of ML;

{ML-Act3} Activities intended to provide students with practical examples of machine learning applied to design;

{ML-Act4} Activities intended to explain the procedure of training a simplified machine learning model;

{ML-Act4.1} Activities intended to set or aligning with a design goal;

{ML-Act4.2} Activities intended to introduce students in familiarising, collecting or creating data;

{ML-Act4.3} Activities intended to provide a step-by-step tool tutorial, such as RunwayML;

{ML-Act5} Activities intended to allow students to individually train a simplified machine learning model for a design goal;

{ML-Act6} Activities intended to provide students with a structured template for process and results documentation;

{ML-Act7} Activities intended to make students explaining and questioning the process.